

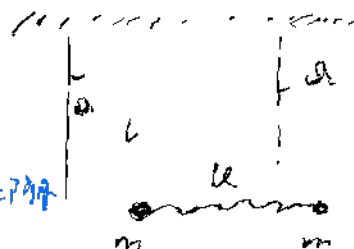
第五章 小振动

§1. 两个角自由度

考虑一个耦合摆:

$$T = \frac{1}{2} m l^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2)$$

$$\Delta y = l(1 - \cos \theta) \approx \frac{l\theta^2}{2} \quad \text{是近似}$$



$$V = -mgl(2 - \cos \theta_1 - \cos \theta_2) + \frac{1}{2} k l^2 (\sin \theta_1 - \sin \theta_2)^2$$

$$\approx \frac{1}{2} mgl(\theta_1^2 + \theta_2^2) + \frac{1}{2} k l^2 (\theta_1 - \theta_2)^2$$

$$L = T - V \quad \begin{cases} \frac{\partial L}{\partial \theta_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} = 0 \\ \frac{\partial L}{\partial \theta_2} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_2} = 0 \end{cases} \Rightarrow \begin{cases} m l \ddot{\theta}_1 + m g \theta_1 + k l (\theta_1 - \theta_2) = 0 \\ m l \ddot{\theta}_2 + m g \theta_2 + k l (\theta_2 - \theta_1) = 0 \end{cases}$$

$$\text{令 } \theta_1 = A \cos(\omega t + \varphi) \quad \theta_2 = B \cos(\omega t + \varphi)$$

$$\Rightarrow \begin{cases} (-\omega^2 m l + m g + k l) A - k l B = 0 \\ (-\omega^2 m l + m g + k l) B - k l A = 0 \end{cases}$$

$$\text{非零解要求} \quad \begin{vmatrix} -\omega^2 m l + m g + k l & -k l \\ -k l & -\omega^2 m l + m g + k l \end{vmatrix} = 0 \Rightarrow \begin{cases} \omega_1 = \sqrt{\frac{g}{l}} \\ \omega_2 = \sqrt{\frac{g}{l}} + \frac{2k}{m} \end{cases}$$

$$\text{特解: } \theta_1 = A_1 \cos(\omega_1 t + \varphi_1), \quad \theta_2 = B_1 \cos(\omega_1 t + \varphi_1) \quad \text{其中 } B_1 = A_1$$

$$\theta_1 = A_2 \cos(\omega_2 t + \varphi_2), \quad \theta_2 = B_2 \cos(\omega_2 t + \varphi_2) \quad \text{其中 } B_2 = -A_2$$

$$\Rightarrow \text{通解: } \theta_1 = \theta_{11} + \theta_{12} = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$$

$$\theta_2 = \theta_{21} + \theta_{22} = A_1 \cos(\omega_1 t + \varphi_1) - A_2 \cos(\omega_2 t + \varphi_2)$$

$$\text{令 } \xi_1 = \theta_1 + \theta_2, \quad \xi_2 = \theta_1 - \theta_2$$

$$\text{特解: } \left. \begin{aligned} \xi_1 &= \theta_{11} + \theta_{21} = 2A_1 \cos(\omega_1 t + \varphi_1) \\ \xi_2 &= \theta_{11} - \theta_{21} = 0 \end{aligned} \right\} \quad \begin{aligned} &\text{简正模(式)} \\ \xi_1 &= \theta_{11} - \theta_{21} = 0 \\ \xi_2 &= \theta_{12} - \theta_{22} = 0 \end{aligned} \right\}$$

$$\text{通解: } \begin{cases} \xi_1 = 2A_1 \cos(\omega_1 t + \varphi_1) \\ \xi_2 = 2A_2 \cos(\omega_2 t + \varphi_2) \end{cases} \Rightarrow \begin{aligned} &\text{简正坐标 } \xi_1, \xi_2 \\ &\Rightarrow \text{简正频率 } \omega_1, \omega_2 \end{aligned}$$

$$* a^2 + b^2 = \frac{1}{2}[(a+b)^2 + (a-b)^2]$$

$$ab = \frac{1}{4}[(a+b)^2 - (a-b)^2]$$

$$T = \frac{1}{2} m l^2 \cdot \frac{1}{2} [(\dot{\theta}_1 + \dot{\theta}_2)^2 + (\dot{\theta}_1 - \dot{\theta}_2)^2] = \frac{1}{4} m l^2 (\dot{\zeta}_1^2 + \dot{\zeta}_2^2)$$

$$V = \frac{1}{2} m g l (\theta_1^2 + \theta_2^2) + \frac{1}{2} k l^2 (\theta_1 - \theta_2)^2$$

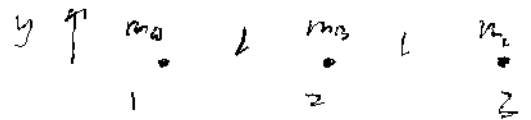
$$= \frac{1}{4} m g l [\zeta_1^2 + \zeta_2^2] + \frac{1}{2} k l^2 \zeta_2^2$$

$$L = \frac{1}{4} m l^2 (\dot{\zeta}_1^2 + \dot{\zeta}_2^2) - \frac{1}{4} m g l (\zeta_1^2 + \zeta_2^2) - \frac{1}{2} k l^2 \zeta_2^2$$

$$\begin{cases} \frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}_1} - \frac{\partial L}{\partial \zeta_1} = 0 \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}_2} - \frac{\partial L}{\partial \zeta_2} = 0 \end{cases} \Rightarrow \begin{cases} 2\ddot{\zeta}_1 + g\zeta_1 = 0 \\ l\ddot{\zeta}_2 + g\zeta_2 + \frac{2kl}{m} \zeta_2 = 0 \end{cases}$$

例：求解双摆运动

§ 2. 三质点振动



1. 横振动 (y)

自由度: y_1, y_2, y_3

$$\text{约束: } \sum_i m_i y_i = 0 \quad \rightarrow \quad y_2 = -\frac{2m_1}{m_3} y_1$$

$$m_1 y_1 + m_3 y_3 = 0 \quad \rightarrow \quad y_3 = -y_1$$

$$V(y_1) = V_0 + \frac{dV}{dy_1}(y_1 - y_{10}) + \frac{1}{2} \frac{d^2V}{dy_1^2}(y_1 - y_{10})^2 \quad \text{忽略高阶项}$$

$$y_1 = y_{10} \text{ 时 } V(y_1) = V_0 + \frac{1}{2} k y_1^2 \quad (k = \frac{d^2V}{dy_1^2} \big|_{y_1=y_{10}})$$

$$T = m_1 \dot{y}_1^2 + \frac{1}{2} m_3 \dot{y}_3^2 = (m_1 + \frac{2m_1^2}{m_3}) \dot{y}_1^2$$

$$L = T - V = (m_1 + \frac{2m_1^2}{m_3}) \dot{y}_1^2 - V_0 - \frac{1}{2} k y_1^2$$

$$\rightarrow \omega = \sqrt{\frac{k m_3}{2m_1(m_1 + 2m_3)}}$$

2. 纵振动 (x)

自由度: x_1, x_2, x_3

$$\text{约束: } m_1 x_1 + m_3 x_2 + m_3 x_3 = 0 \quad \rightarrow \quad x_2 = -\frac{m_1}{m_3} (x_1 + x_3)$$

$$T = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{x}_3^2) + \frac{1}{2} m_3 \dot{x}_2^2 = \frac{1}{2} (m_1 + \frac{m_1^2}{m_3}) (\dot{x}_1^2 + \dot{x}_3^2) + \frac{m_1^2}{2m_3} \dot{x}_1 \dot{x}_3$$

$$V = \frac{1}{2} k (x_2 - x_1)^2 + \frac{1}{2} k (x_3 - x_2)^2$$

$$= \frac{k}{2m_3} \{ [(m_1 + m_3) x_1 + m_1 x_3]^2 + [m_1 x_1 + (m_1 + m_3) x_3]^2 \}$$

$$= \frac{k}{2m_3} [(2m_1^2 + 2m_1 m_3 + m_3^2) (x_1^2 + x_3^2) + 4m_1(m_1 + m_3) x_1 x_3]$$

$$L = \frac{1}{2} (m_1 + \frac{m_1^2}{m_3}) (\dot{x}_1^2 + \dot{x}_3^2) + \frac{m_1^2}{2m_3} \dot{x}_1 \dot{x}_3$$

$$- \frac{k}{2m_3} [(2m_1^2 + 2m_1 m_3 + m_3^2) (x_1^2 + x_3^2) + 4m_1(m_1 + m_3) x_1 x_3]$$

$$\begin{cases} \frac{\partial}{\partial c} \frac{\partial L}{\partial x_i} - \frac{\partial L}{\partial x_i} = 0 \\ \frac{\partial}{\partial c} \frac{\partial L}{\partial x_j} - \frac{\partial L}{\partial x_j} = 0 \end{cases}$$

§3. 小振动的一般理论

1. 选取5个广义坐标 q_α ($\alpha=1, \dots, 5$), 平衡位置 $q_{\alpha 0} = 0$

$$V = V(q_\alpha)$$

$$= \underbrace{V_0}_{=0} + \sum_{\alpha} \left(\frac{\partial V}{\partial q_{\alpha 0}} \right) q_{\alpha} + \frac{1}{2} \sum_{\alpha} \sum_{\beta} \left(\frac{\partial^2 V}{\partial q_{\alpha 0} \partial q_{\beta 0}} \right) q_{\alpha} q_{\beta}$$

刚度系数 $= k_{\alpha\beta} = k_{\beta\alpha} = \left(\frac{\partial^2 V}{\partial q_{\alpha 0} \partial q_{\beta 0}} \right)_{\alpha}$

$$V = \frac{1}{2} \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} q_{\alpha} q_{\beta}$$

定常约束下: $T = \frac{1}{2} \sum_i m_i (\dot{\vec{r}}_i \cdot \dot{\vec{r}}_i)$

$$= \frac{1}{2} \sum_i \sum_{\alpha} \sum_{\beta} m_i \left(\frac{\partial \vec{r}_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \right) \cdot \left(\frac{\partial \vec{r}_i}{\partial q_{\beta}} \dot{q}_{\beta} \right)$$

惯性系数: $m_{\alpha\beta} = \sum_i m_i \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \cdot \frac{\partial \vec{r}_i}{\partial q_{\beta}}$

$$T = \frac{1}{2} \sum_{\alpha} \sum_{\beta} m_{\alpha\beta} \dot{q}_{\alpha} \dot{q}_{\beta}$$

$$L = \frac{1}{2} \sum_{\alpha} \sum_{\beta} (m_{\alpha\beta} \dot{q}_{\alpha} \dot{q}_{\beta} - k_{\alpha\beta} q_{\alpha} q_{\beta})$$

2. 化简求和

3. 利用正则格朗日方程求解.

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} \left(\frac{1}{2} \sum_p \sum_q m_{pq} \dot{q}_p \dot{q}_q \right) + \frac{\partial L}{\partial q_\alpha} \left(\frac{1}{2} \sum_p \sum_q k_{pq} q_p q_q \right) = 0$$

$$\Rightarrow \frac{d}{dt} \left(\sum_p m_{\alpha p} \dot{q}_p \right) + \sum_q k_{\alpha q} q_q = 0$$

$$\Rightarrow \sum_p (m_{\alpha p} \ddot{q}_p + k_{\alpha p} q_p) = 0$$

$$\text{令 } q_p = A_p e^{i\omega t} \Rightarrow \sum_p (\lambda^2 m_{\alpha p} + k_{\alpha p}) A_p e^{i\omega t} = 0 \quad \alpha = 1 \cdots S$$

$$A \text{ 有非零解} \Rightarrow |\{C_{\alpha p}\}|_{S \times S} = |\{\lambda^2 m_{\alpha p} + k_{\alpha p}\}|_{S \times S} = 0$$

$$\text{设 } S \text{ 个 } \lambda^2 \text{ 的解为 } \lambda_l^2 = -\omega_l^2 \quad (l = 1 \cdots S) \quad (\lambda \text{ 有 } 2S \text{ 个})$$

* 对于 $\lambda^2 = \lambda_l^2$ 这是一个解 (给简正模), 代入

$$\text{在 } \sum_p (m_{\alpha p} \lambda_l^2 + k_{\alpha p}) A_p^l = 0 \quad (\alpha = 1, 2, \dots, S) \text{ 中, } S-1 \text{ 个方程独立}$$

$$\Rightarrow \sum_{\alpha=1}^{S-1} (m_{\alpha p} \lambda_l^2 + k_{\alpha p}) A_p^l = -(m_{S p} \lambda_l^2 + k_{S p}) A_S^l$$

$$\Rightarrow \sum_{\alpha=1}^{S-1} \sum_{p=1}^S (m_{\alpha p} \lambda_l^2 + k_{\alpha p}) A_p^l = - \sum_{\alpha=1}^{S-1} (m_{\alpha S} \lambda_l^2 + k_{\alpha S}) A_S^l$$

$$\xrightarrow{\alpha \downarrow} \begin{bmatrix} m_{11} \lambda_l^2 + k_{11} & \cdots & m_{1,S-1} \lambda_l^2 + k_{1,S-1} \\ \vdots & & \vdots \\ \vdots & & \vdots \\ m_{S-1,1} \lambda_l^2 + k_{S-1,1} & \cdots & m_{S-1,S-1} \lambda_l^2 + k_{S-1,S-1} \end{bmatrix}_{(S-1) \times (S-1)} \begin{bmatrix} A_1^l \\ A_2^l \\ \vdots \\ A_{S-1}^l \end{bmatrix}_{(S-1) \times 1} = \begin{bmatrix} -(m_{1S} \lambda_l^2 + k_{1S}) A_S^l \\ \vdots \\ -(m_{S-1,S} \lambda_l^2 + k_{S-1,S}) A_S^l \end{bmatrix}_{(S-1) \times 1}$$

$$A_p^l = - \sum_{\alpha=1}^{S-1} (m_{\alpha S} \lambda_l^2 + k_{\alpha S}) A_S^l \cdot \Delta_{\alpha p} \quad (\text{Cramer's rule}) \quad \text{其中 } \Delta_{\alpha p} \text{ 为 } D \text{ 中 } \Delta_{\alpha p} \text{ 的代数余子式}$$

$$\text{对 } \lambda_l^2 \text{ 的 } \{C_{\alpha p}\}_{S \times S} = \{\lambda_l^2 m_{\alpha p} + k_{\alpha p}\}_{S \times S}, \quad (-1)^{S+S} = 1 \quad |D_{SS}| = |M_{SS}|$$

$$\Delta = \Delta_{SS}^l \Rightarrow \text{对于 } \lambda^2 = \lambda_l^2 \text{ 的, 正则格朗日方程 } C \text{ 中 } S \text{ 个元基的(代数)余子式}$$

$$-\sum_{\alpha=1}^s (m_{\alpha} \lambda_{\alpha}^2 + k_{\alpha} s) \alpha_{\alpha\beta} = \alpha_{\alpha\beta}^1 \cdot \text{如何确定 } (-1)^{\beta+s} \rightarrow (-1) \cdot M_{\alpha\beta}?$$

$$\Rightarrow A_{\beta}^1 = \frac{\alpha_{\alpha\beta}^1}{\alpha_{\alpha\beta}^1} A_{\alpha}^1 \quad \beta=s \text{ 吗?} \rightarrow \beta=s \quad A_{\alpha}^1 = A_{\beta}^1, \text{ 自然成立}$$

对于“一维函数” $\rightarrow$ 一套 $A_1^1, A_2^1, \dots, A_s^1$

* 注意: 此时 λ 有两个 (λ_1 和 $-\lambda_1$)

特解 q_k 是两解的线性叠加 $A_{\beta 1}^1 = A_{\beta}^1 \cdot A_1, \quad A_{\beta 2}^1 = A_{\beta}^1 \cdot A_2$

$\lambda^2 = \lambda_1^2$ 下

$$\begin{aligned} \text{特解: } q_k^1 &= A_{\beta}^1 e^{\lambda_1 t} = A_{\beta 1}^1 e^{\lambda_1 t} + A_{\beta 2}^1 e^{-\lambda_1 t} \\ &= A_{\beta}^1 (A_1 e^{i\omega_1 t} + A_2 e^{-i\omega_1 t}) \\ &= \alpha_{\alpha\beta}^1 (A_1 e^{i\omega_1 t} + A_2 e^{-i\omega_1 t}) = C^1 \alpha_{\alpha\beta}^1 \cos(\omega_1 t + \varphi_1) \end{aligned}$$

$$\text{通解 } q_k = \sum_{i=1}^s q_k^i = \sum_{i=1}^s \alpha_{\alpha\beta}^i \tilde{z}_i$$

$$\text{其中 } \tilde{z}_i = C^i \cos(\omega_i t + \varphi_i)$$

$\Rightarrow$ 所有 $\gamma \times$ 全都有相同“频率”? $\omega_1, \omega_2, \dots, \omega_s$

4. 证明 $\lambda^2 < 0$

$$\sum_{\beta} (\lambda^2 m_{\alpha\beta} + k_{\alpha\beta}) A_{\beta} = 0$$

$$\Rightarrow \sum_{\alpha} \sum_{\beta} (\lambda^2 m_{\alpha\beta} + k_{\alpha\beta}) A_{\alpha}^* A_{\beta} = 0$$

$$\lambda^2 = - \frac{\sum_{\alpha} \sum_{\beta} k_{\alpha\beta} A_{\alpha}^* A_{\beta}}{\sum_{\alpha} \sum_{\beta} m_{\alpha\beta} A_{\alpha}^* A_{\beta}}$$

$$A_{\alpha} = a_{\alpha} + i b_{\alpha}$$

$$A_{\beta} = a_{\beta} + i b_{\beta}$$

$$\Rightarrow \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} A_{\alpha}^* A_{\beta}$$

$$= \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} (a_{\alpha} b_{\alpha} + b_{\alpha} b_{\beta}) + i \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} (a_{\alpha} b_{\beta} - a_{\beta} b_{\alpha})$$

$$= \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} (a_{\alpha} b_{\alpha} + b_{\alpha} b_{\beta}) \quad \Rightarrow > 0$$

$$\text{对于任意实数 } a, b, V = \frac{1}{2} \sum_{\alpha} \sum_{\beta} \left(\frac{\partial^2 V}{\partial q_{\alpha} \partial q_{\beta}} \right) q_{\alpha} q_{\beta} = \frac{1}{2} \sum_{\alpha} \sum_{\beta} k_{\alpha\beta} q_{\alpha} q_{\beta} > 0$$

$$= \frac{1}{2} \vec{q}^T K \vec{q} \quad \text{且 } K \text{ 正定} \quad \text{且 } \vec{q} \neq 0 \Rightarrow \vec{q}^T K \vec{q} > 0$$

$$\Rightarrow \sum_{\alpha} \sum_{\beta} m_{\alpha\beta} A_{\alpha}^* A_{\beta} > 0 \quad \text{略证}$$

5. 矩阵描述

定常约束: $\vec{r}_i = \vec{r}_i(q_1, \dots, q_s)$ 则 T 是 q 的正定二次型.

$$T = \sum_i \frac{1}{2} m_i \dot{r}_i^2 = \frac{1}{2} \sum_{\alpha, \beta} m_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta$$

$$\vec{Q} = \begin{bmatrix} q_1 \\ \vdots \\ q_s \end{bmatrix} \quad \dot{\vec{Q}} = \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_s \end{bmatrix}$$

$$T = \frac{1}{2} \dot{\vec{Q}}^T M \dot{\vec{Q}} \quad , \quad V = \frac{1}{2} \dot{\vec{Q}}^T K \dot{\vec{Q}}$$

M : 惯性矩阵 正定

K : 刚度矩阵 稳定平衡附近正定(或半正定); $\forall \vec{x} \in \mathbb{R}^n$ 有 $\vec{x}^T A \vec{x} > 0$ (≥ 0)
(均为实对称矩阵)

$$L = \frac{1}{2} \dot{\vec{Q}}^T M \dot{\vec{Q}} - \frac{1}{2} \dot{\vec{Q}}^T K \dot{\vec{Q}}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\vec{Q}}} - \frac{\partial L}{\partial \vec{Q}} = 0, \quad M \ddot{\vec{Q}} + K \dot{\vec{Q}} = 0$$

$$dL = \frac{1}{2} d \left[\sum_{\alpha} \sum_{\beta} (m_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta - k_{\alpha\beta} q_\alpha q_\beta) \right]$$

$$= \frac{1}{2} \sum_{\alpha} \sum_{\beta} [m_{\alpha\beta} (\dot{q}_\alpha d\dot{q}_\beta + \dot{q}_\beta d\dot{q}_\alpha) - k_{\alpha\beta} (q_\alpha dq_\beta + q_\beta dq_\alpha)]$$

$$= \sum_{\alpha} \sum_{\beta} m_{\alpha\beta} \dot{q}_\alpha d\dot{q}_\beta - k_{\alpha\beta} q_\alpha dq_\beta$$

$$\Rightarrow \frac{\partial L}{\partial \dot{\vec{Q}}} = \left\langle \frac{\partial L}{\partial \dot{q}_1}, \dots, \frac{\partial L}{\partial \dot{q}_s} \right\rangle = \left(\sum_{\beta} \sum_{\alpha} m_{\alpha\beta} \dot{q}_\alpha, -k_{\alpha\beta} q_\alpha \right)$$

$$\text{选取简正坐标后, } \dot{\vec{Q}} = \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_s \end{bmatrix} = \begin{bmatrix} \dot{A}_1 \\ \vdots \\ \dot{A}_s \end{bmatrix} = \dot{\vec{A}} \}$$

$$M \dot{\vec{A}} + K \vec{A} = 0, \quad \ddot{\vec{A}} + \omega^2 \vec{A} = 0$$

$$\Rightarrow [-\omega^2 M + K] \vec{A} = 0 \quad \Rightarrow (-\omega^2 M + K) \vec{A} = 0$$

M 为 $n \times n$ 矩阵

$M^{-1}KA = \omega^2 A$ (A 是变换 $(M^{-1}K)$ 的一个特征向量, 特征值为 ω^2)

A 存在非零解 $\rightarrow$ 特征方程: $| -\omega^2 M + K | = 0$

(M, K 为 $n \times n$ 矩阵)

关于 ω^2 的 s 次方程, 解为 $\omega_1^2, \dots, \omega_s^2$

取单根 ω_i^2 代入 $(-\omega_i^2 M + K) A = 0$ (此时 $s-1$ 个方程独立)

解得比值: $A_1^i : A_2^i : \dots : A_s^i$

$$q_1 = A_1^i z_1 \quad \dots \quad q_s = A_s^i z_i$$

对于 r 重根 ω_m^2 , 代入 $(-\omega_m^2 M + K) A$

可确定 r 种 A 比值 $A_1^m : A_2^m : \dots : A_s^m$

$\hookrightarrow r$ 种简正模式

$$\Rightarrow \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_s \end{pmatrix} = \begin{pmatrix} A_1^1 & A_1^2 & \dots & A_1^s \\ A_2^1 & A_2^2 & \dots & A_2^s \\ \vdots & \vdots & \ddots & \vdots \\ A_s^1 & A_s^2 & \dots & A_s^s \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_s \end{pmatrix}$$

$$\vec{Q} = A^T \vec{z}$$

A^i 中每一个列向量 $\vec{A}^i = \begin{pmatrix} A_1^i \\ A_2^i \\ \vdots \\ A_s^i \end{pmatrix}$ 都是 $(M^{-1}K)$ 特征值为 ω_i^2 的特征向量

$$(-\omega_i^2 M + K) \vec{A}^i = 0$$

$$\text{同乘 } A^m \rightarrow \omega_i^2 \tilde{A}^m M A^i = \tilde{A}^m K A^i \quad (A)$$

$$\text{变换 } m, i \quad \omega_m^2 \tilde{A}^m M A^m = \tilde{A}^m K A^m$$

取位置 $\rightarrow \omega_m^2 \tilde{A}^m M A^l = \tilde{A}^m K A^l$ 由 M, K 为实对称矩阵
 $\tilde{M} = M, \tilde{K} = K$

(a) - (b):

$$(\omega_l^2 - \omega_m^2) \tilde{A}^m M A^l = 0$$

① 当 ω_l^2 为单根:

$$\tilde{A}^m M A^l = 0 \quad (m \neq l) \quad (c) \Rightarrow \text{特征向量 } A^m, A^l \text{ 彼此正交}$$

$$M \text{ 是正定的} \rightarrow \tilde{A}^l M A^l = \mu^l > 0$$

可定义归一化的特征向量: $A^l \rightarrow \frac{1}{\sqrt{\mu^l}} A^l$

正交归一化条件: $\tilde{A}^m M A^l = \mu^l \delta_{lm}$ 其中 $\delta_{lm} = \begin{cases} 1 & l=m \\ 0 & l \neq m \end{cases}$

② 当 ω_l^2 为 r 重根:

有 $A_1, A_2, \dots, A_r$ 个特征向量, 类似, 与其它特征向量正交:

$$(\omega_l^2 - \omega_k^2) \tilde{A}^l M A^k = 0 \quad (l \notin [1, r], k \in [1, r])$$

格拉姆-施密特方法:

构造 $\{A_1', \dots, A_r'\}$ 两两正交且与其它特征向量正交:

在简正坐标: $Q = A' \xi$ 下, T, V 化为二次型

$$\begin{aligned} T &= \frac{1}{2} \dot{Q}^T M \dot{Q} = \frac{1}{2} \underbrace{\{\tilde{A}_1' \dots \tilde{A}_r'\}}_{\left\{ \tilde{A}_1' \dots \tilde{A}_r' \right\}} M \underbrace{\{A_1' \dots A_r'\}}_{\left\{ A_1' \dots A_r' \right\}} \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_r \end{bmatrix} \\ &= \frac{1}{2} \sum_{l,m} \xi_l \tilde{A}_l'^T M A_m' \xi_m \quad \leftarrow \left(\xi_1 \tilde{A}_1' \right) M \left(\xi_r \tilde{A}_r' \right) \\ &= \frac{1}{2} \sum_l \mu^l \xi_l^2 = \frac{1}{2} \sum_l \mu^l \xi_l^2 \end{aligned}$$

$$V = \frac{1}{2} Q^T K Q = \frac{1}{2} \{ \tilde{A}_1' \dots \tilde{A}_r' \} K \{ A_1' \dots A_r' \}$$

$$= \frac{1}{2} \sum_{l,m} \xi_l \tilde{A}_l'^T K A_m' \xi_m = \frac{1}{2} \sum_{l,m} \xi_l \tilde{A}_l'^T \omega_l^2 M A_m' \xi_m$$

$$= \frac{1}{2} \sum_{l,m} \omega_l^2 \mu^l \{ \}^1 \{ \}^m = \frac{1}{2} \sum_l \omega_l^2 \mu^l \{ \}^2$$

$$\Rightarrow M = \text{diag}(\mu^1, \dots, \mu^N)$$

$$K = \text{diag}(\omega_1^2 \mu^1, \dots, \omega_N^2 \mu^N)$$

例：用矩阵理论求解双摆问题

$$L = T - V$$

$$= \frac{1}{2} m_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 [l_1 \dot{\theta}_1 + l_2 \dot{\theta}_2]^2 + m_1 g l_1 (1 - \frac{\theta_1^2}{2})$$

$$+ m_2 g [l_1 (1 - \frac{\theta_1^2}{2}) + l_2 (1 - \frac{\theta_2^2}{2})]$$

$$= \frac{1}{2} m_1 \dot{\theta}_1^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2$$

$$- (m_1 + m_2) g l_1 \frac{\theta_1^2}{2} - m_2 g l_2 \frac{\theta_2^2}{2}$$

$$= \frac{1}{2} \dot{\mathbf{Q}}^T M \dot{\mathbf{Q}} - \frac{1}{2} \mathbf{Q}^T K \mathbf{Q}$$

其中：

$$\mathbf{Q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad M = \begin{bmatrix} m_1 l_1^2 & m_2 l_1 l_2 \\ m_2 l_1 l_2 & m_2 l_2^2 \end{bmatrix} \quad K = \begin{bmatrix} (m_1 + m_2) g l_1 & 0 \\ 0 & m_2 g l_2 \end{bmatrix}$$

$$L = \frac{1}{2} \dot{\mathbf{Q}}^T M \dot{\mathbf{Q}} - \frac{1}{2} \mathbf{Q}^T K \mathbf{Q}$$

$$\text{拉格朗日方程：} \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{Q}}} - \frac{\partial L}{\partial \mathbf{Q}} = M \ddot{\mathbf{Q}} + K \mathbf{Q} = 0$$

$$\text{广义坐标：} \mathbf{Q} = A \mathbf{z}, \quad \ddot{\mathbf{z}} + \tilde{\omega}^2 \mathbf{z} = 0$$

$$(-\omega^2 M + K) A \mathbf{z} = 0$$

$$\text{非平凡解：} |-\omega^2 M + K| = 0$$

$$\Leftrightarrow \begin{aligned} (m_1 + m_2) g l_1 - m_1 \omega^2 l_1^2 &= m_2 l_1 l_2 \omega^2 \\ -m_2 l_1 l_2 \omega^2 &= m_2 g l_2 - m_2 \omega^2 l_2^2 \end{aligned}$$

$$= m_1 m_2 l_1^2 l_2^2 \omega^4 - [(m_1 + m_2) m_2 g l_1 l_2^2 + m_1 m_2 g l_1^2 l_2] \omega^2$$

$$+ (m_1 + m_2) m_2 g^2 l_1 l_2 - m_2^2 l_1^2 l_2^2 \omega^4 = 0$$

$$\Rightarrow \omega_1 = \sqrt{2 + \sqrt{2}} \frac{g}{l_1}, \quad \omega_2 = \sqrt{2 - \sqrt{2}} \frac{g}{l_1}$$

求解相应特征向量： $\mathbf{A}^{(1)}, \mathbf{A}^{(2)}$

$$\mathbf{A}^{(1)}: \mathbf{A}^{(1)}_2 = \quad, \quad \mathbf{A}^{(2)}: \mathbf{A}^{(2)}_2 = \quad$$